



Modelling Financial Markets

Seminar 11: Continuous-Time Models — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- BET: Bucharest Exchange Trading (index)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CI: Confidence Interval
- CIR: Cox–Ingersoll–Ross (model)
- DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- FRED: Federal Reserve Economic Data
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GBM: Geometric Brownian Motion
- LR: Likelihood Ratio (test)
- MLE: Maximum Likelihood Estimation
- OLS: Ordinary Least Squares
- OU: Ornstein–Uhlenbeck (process)
- US: United States
- VIX: CBOE Volatility Index
- XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Introduction

Part A: Pen and Paper

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Apply Itô's lemma and compute the moments of Brownian functionals step by step
- ▣ Map between continuous-time models and their discrete-time versions (AR(1), Poisson mixtures)
- ▣ Estimate GBM, Vasicek, Merton and Heston parameters with standard errors
- ▣ Recognise when standard inference fails: biased $\hat{\kappa}$, boundary likelihood-ratio tests

- ▣ Background: Lecture 11; textbooks [Franke, Härdle & Hafner \(2019\)](#) and [Borak, Härdle & López-Cabrera \(2013\)](#)

Structure

- ▣ **Part A:** eight pen-and-paper exercises (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Time in years, daily step $\Delta t = 1/252$ (Bitcoin 1/365); log returns; drifts and volatilities per year
- Report every estimate with its standard error or confidence interval (CI)

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (Itô's lemma for $\ln S$), A3 (quadratic variation), A5 (Vasicek moments), A7 (Merton cumulants)
- Part B: B1 (orders of convergence), B2 (GBM on the S&P 500), B4 (Vasicek and the bias of $\hat{\kappa}$), B6 (Merton and a bootstrap LR test)

Proposed: you solve them

- Part A: A2 (Siegel's paradox; model: A1), A4 (the Itô integral; model: A3), A6 (AR(1) to OU; model: A5), A8 (rare large jumps; model: A7)
- Part B: B3 (GBM on BET and Bitcoin; model: B2), B5 (OU for the VIX; model: B4), B7 (jumps in Bitcoin; model: B6), B8 (Heston from the VIX; model: B4, B6)
- Part C: C1, open question and project seed; C2, critique of an AI answer

Setup: Python Environment and Data

-  **Open in Colab:** `chapter11_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `matplotlib`
- Series: S&P 500 (1990–2026), BET (2000–2026), Bitcoin (2014–2026), VIX (1990–2026); daily closes up to 18 September 2026
- 3-month US Treasury bill rate (FRED series DTB3), 1954–2026
- Equity indices: weekdays only, days with an unchanged close removed; Bitcoin: 7 days a week

Formulas Used Today

- ▣ Itô's lemma: $dX = a dt + b dW \Rightarrow df(X) = (af' + \frac{1}{2}b^2f'')dt + bf' dW$
- ▣ GBM: $\ln(S_T/S_0) \sim N((\mu - \sigma^2/2)T, \sigma^2 T)$; $\mathbb{E}[S_T] = S_0 e^{\mu T}$
- ▣ OU: $\mathbb{E}[x_t | x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$, $\text{Var}(x_t | x_0) = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$; AR(1): $b = e^{-\kappa \Delta t}$
- ▣ Merton cumulants: $\kappa_2 = \sigma^2 \Delta t + \lambda \Delta t(\mu_J^2 + \sigma_J^2)$, $\kappa_3 = \lambda \Delta t(\mu_J^3 + 3\mu_J \sigma_J^2)$, $\kappa_4 = \lambda \Delta t(\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$
- ▣ Heston: $dv = \kappa(\theta - v)dt + \xi\sqrt{v}dW^{(2)}$, $\text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$; Feller: $2\kappa\theta \geq \xi^2$

A1: Itô's Lemma for $\ln S$ [Solved]

Task

- A stock follows GBM with $\mu = 8\%$, $\sigma = 20\%$ per year
- (a) Derive $d \ln S_t$ with Itô's lemma
- (b) Mean and median of S_{10}/S_0
- (c) Probability of a loss after 10 years; probability of ending below the mean

Solution

- $f = \ln x$, $f' = 1/x$, $f'' = -1/x^2$:
 $d \ln S = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW$, log drift
 $m = 0.08 - 0.02 = 6\%$
- $\mathbb{E}[S_{10}/S_0] = e^{0.6} = 2.2255$; median $e^{0.6} = 1.8221$
- $P(S_{10} < S_0) = \Phi\left(-\frac{0.06\sqrt{10}}{0.2}\right) = \Phi(-0.9487) = 17.14\%$
- $P(S_{10} < \mathbb{E}[S_{10}]) = \Phi(\sigma\sqrt{T}/2) = \Phi(0.316) = 62.4\%$
- Six investors in ten end below the advertised mean



A2: Siegel's Paradox [Proposed]

Solution discussed in class

Task

- The EUR/RON rate X_t follows GBM with $\mu = 2\%$, $\sigma = 4.4\%$ per year
- (a) Apply Itô's lemma to $Y_t = 1/X_t$ (RON/EUR)
- (b) Expected percentage drift of Y : is it -2% ?
- (c) Compare the log drifts of X and Y
- Model: A1

A3: Quadratic Variation of Brownian Motion [Solved]

Task

- ▣ $Q_n = \sum_{k=1}^n (\Delta W_k)^2$ on $[0, T]$ with n equal steps
- ▣ (a) $\mathbb{E}[Q_n]$ and $\text{Var}(Q_n)$
- ▣ (b) Numbers for $T = 1$, $n = 252$
- ▣ (c) Mean total variation $\mathbb{E} \sum |\Delta W_k|$ for the same grid

Solution

- ▣ $\Delta W_k = \sqrt{\Delta t} Z_k$: $\mathbb{E}[(\Delta W_k)^2] = \Delta t$,
 $\text{Var}((\Delta W_k)^2) = \Delta t^2 \text{Var}(Z^2) = 2\Delta t^2$
- ▣ $\mathbb{E}[Q_n] = T$, $\text{Var}(Q_n) = 2T^2/n$ (independent terms)
- ▣ $T = 1$, $n = 252$: $\text{Var}(Q_n) = 0.00794$, standard deviation 0.0891
- ▣ $\mathbb{E}|\Delta W_k| = \sqrt{\Delta t} \sqrt{2/\pi} \Rightarrow$
 $\mathbb{E} \sum |\Delta W_k| = \sqrt{2nT/\pi} = 12.67$
- ▣ As $n \rightarrow \infty$: $Q_n \rightarrow T$ in mean square, the total variation diverges

A4: The Itô Integral $\int_0^T W \, dW$ [Proposed]

Solution discussed in class

Task

- ▣ (a) Show with Itô's lemma for $f(x) = x^2$ that $\int_0^T W \, dW = \frac{1}{2}(W_T^2 - T)$
- ▣ (b) Its mean and variance for $T = 1$; check the variance with the Itô isometry
- ▣ (c) The Stratonovich integral equals $\frac{1}{2} W_T^2$: what is its mean?
- ▣ Model: A3

A5: Vasicek, Step by Step [Solved]

Task

- Short rate $dr = \kappa(\theta - r)dt + \sigma dW$ with $\kappa = 0.5$, $\theta = 3\%$, $\sigma = 1\%$; today $r_0 = 6\%$ Vasicek (1977)
- (a) Half-life of a deviation
- (b) Mean and standard deviation of r_1 given r_0
- (c) Long-run probability of a negative rate

Solution

- $\ln 2/0.5 = 1.386$ years
- $e^{-0.5} = 0.6065$: $\mathbb{E}[r_1] = 3\% + 3\% \cdot 0.6065 = 4.82\%$
- $\text{Var}(r_1) = \frac{0.01^2}{2 \cdot 0.5} (1 - e^{-1}) \Rightarrow \text{s.d. } 0.795\%$
- Stationary: $N(3\%, (0.01/\sqrt{1})^2) = N(3\%, 1\%^2)$;
 $P(r < 0) = \Phi(-3) = 0.135\%$
- Small but not zero: Vasicek allows negative rates, CIR does not

A6: From an AR(1) to an OU Process [Proposed]

Solution discussed in class

Task

- An AR(1) on monthly data ($\Delta t = 1/12$): $x_{t+1} = 0.0012 + 0.97 x_t + \varepsilon_t$, $\hat{\sigma}_\varepsilon = 0.0025$
- (a) Recover κ , θ , σ of the OU process
- (b) Half-life in years and the stationary standard deviation
- (c) Would the same κ give $b = 0.97$ with daily data?
- Model: A5

A7: Cumulants of the Merton Return [Solved]

Task

- Merton model with $\sigma = 15\%$, $\lambda = 5$ jumps per year, $\mu_J = -3\%$, $\sigma_J = 4\%$ [Merton \(1976\)](#)
- (a) Daily variance and the share due to jumps ($\Delta t = 1/252$)
- (b) Daily skewness and excess kurtosis
- (c) Probability of at least one jump in a day and in a year

Solution

- $\sigma^2 \Delta t = 0.893 \times 10^{-4}$, $\lambda \Delta t (\mu_J^2 + \sigma_J^2) = 0.496 \times 10^{-4}$:
 $\kappa_2 = 1.389 \times 10^{-4}$ (s.d. 1.18%), jumps 35.7%
- $\kappa_3 = -3.393 \times 10^{-6}$, $\kappa_4 = 3.399 \times 10^{-7}$
- Skewness $\kappa_3 / \kappa_2^{3/2} = -2.07$, excess kurtosis $\kappa_4 / \kappa_2^2 = 17.6$
- $1 - e^{-5/252} = 1.96\%$ per day; $1 - e^{-5} = 99.3\%$ per year

A8: Rare, Large Jumps [Proposed]

Solution discussed in class

Task

- ▣ Same $\sigma = 15\%$, but crashes: $\lambda = 0.5$ per year, $\mu_J = -10\%$, $\sigma_J = 5\%$
- ▣ (a) Daily skewness and excess kurtosis
- ▣ (b) Probability of observing no jump in 1 year and in 10 years
- ▣ (c) Why is λ hard to estimate here, and what does the lecture MLE ($\hat{\lambda} \approx 100$) capture instead?
- ▣ Model: A7

B1: Orders of Convergence [Solved]

Task

- Test equation $dX = 2X dt + X dW$, $X_0 = 1$, $T = 1$ [Higham \(2001\)](#); 20,000 paths
- (a) Strong errors of Euler–Maruyama and Milstein for $\Delta t = 2^{-11}, \dots, 2^{-5}$
- (b) Regress $\ln(\text{error})$ on $\ln \Delta t$: slope with 95% CI
- (c) Weak error of Euler–Maruyama: exact and by Monte Carlo

Interpretation

- Are the slopes consistent with the theoretical orders $1/2$ and 1 ?
- Why does the Monte Carlo weak error give a wrong slope?
- How many paths would you need to see the weak error at the finest step?

B1: Results [Solved]

	Slope	s.e.	95% CI
Strong, Euler–Maruyama	0.518	0.004	[0.509, 0.528]
Strong, Milstein	0.986	0.003	[0.977, 0.995]
Weak, Euler–Maruyama, exact mean	0.985	0.004	[0.976, 0.994]
Weak, Euler–Maruyama, Monte Carlo	0.511	0.052	[0.376, 0.646]

- Strong slopes close to 0.5 and 1; the tiny s.e. comes from common random numbers and ignores the curvature at large steps
- Monte Carlo: s.d. of X_T is 9.69, so the standard error of the mean is 0.068, ten times the weak error at $\Delta t = 2^{-11}$ (0.0072)
- To resolve it at the 95% level: $(1.96 \cdot 9.69/0.0072)^2 \approx 6\,937\,381$ paths

 MFM_ch11_convergence

B2: GBM on the S&P 500 [Solved]

Task

- Daily log returns 1990–2026
- (a) MLE of σ , of the log drift m and of μ , with standard errors and a 95% CI for m
- (b) Test the GBM implications: Jarque–Bera for Normality, Ljung–Box on $|r_t|$ (20 lags) for independence
- (c) Variance ratio of 20-day to 1-day returns (GBM: 1)

Interpretation

- Which parameter is known well and which one badly? Why?
- Which GBM assumption does each test reject?
- Can you still use $\hat{\sigma}$ for option pricing?

B2: Results [Solved]

- $n = 9\,240$ days, $T = 36.7$ years: $\hat{\sigma} = 18.0\%$ (s.e. 0.13), $\hat{m} = 8.3\%$ (s.e. 3.0), $\hat{\mu} = 10.0\%$
 - ▶ 95% CI for m : [2.5%, 14.2%]; ± 1 point would need 1 244 years
- Jarque–Bera 45 718 (excess kurtosis 10.9): Normality rejected
 - ▶ Ljung–Box on $|r_t|$: 13 389 against the 5% critical value 31.4; ACF at lag 1 is 0.27: independence rejected
- Variance ratio 0.80: 20-day variance below 20 times the daily one (mild mean reversion of daily moves)
- Interpretation: σ is estimable and useful, but it is an **average** of a volatility that changes over time [Cont \(2001\)](#)

B3: GBM on BET and Bitcoin [Proposed]

Solution discussed in class

Task

- Repeat B2 for BET (2000–2026, $\Delta t = 1/252$) and Bitcoin (2014–2026, $\Delta t = 1/365$)
- (a) $\hat{\sigma}$, $\hat{m}$ with 95% CIs; years of data needed for ± 1 point
- (b) Jarque–Bera, Ljung–Box on $|r_t|$, variance ratio
- Model: B2

Interpretation

- Is the drift of Bitcoin significantly positive?
- Which market is further from GBM, and in which dimension?
- What does a variance ratio above 1 for BET suggest?

B4: Vasicek and the Bias of $\hat{\kappa}$ [Solved]

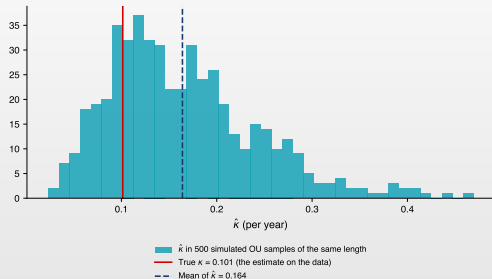
Task

- ▣ 3-month Treasury bill rate (DTB3), 1954–2026, $\Delta t = 1/252$ [Vasicek \(1977\)](#)
- ▣ (a) Exact AR(1) estimation of κ , θ , σ ; delta-method standard errors; 95% CI for κ
- ▣ (b) Simulate 500 OU paths of the same length with the estimates as the truth; re-estimate κ on each
- ▣ (c) Bias, bias-corrected κ and half-life

Interpretation

- ▣ Is mean reversion significant?
- ▣ Does the simulated bias match the $4/T$ rule [Tang & Chen \(2009\)](#)?
- ▣ What does the bias do to hedging or pricing with Vasicek?

B4: Results [Solved]



- $n = 18\,170$ days (72.1 years): $\hat{b} = 0.99960$, $\hat{\kappa} = 0.101$ (s.e. 0.051), 95% CI [0.001, 0.202]
 - ▶ $\hat{\theta} = 4.55\%$ (s.e. 1.55), $\hat{\sigma} = 1.33\%$, half-life 6.8 years
- Simulation: mean $\hat{\kappa} = 0.164$ (s.d. 0.078); bias 0.063 against $4/T = 0.055$; 78% of estimates above the truth
 - ▶ Bias-corrected $\kappa = 0.101 - 0.063 = 0.039$: half-life 17.8 years instead of 6.8
- The CI barely excludes zero, and after bias correction the rate is close to a random walk over 72 years

B5: Mean Reversion of the VIX [Proposed]

Solution discussed in class

Task

- ▣ $x_t = \ln VIX_t$, 1990–2026
- ▣ (a) OU estimates; half-life in trading days with a 95% CI (from the CI of κ)
- ▣ (b) Simulated bias of $\hat{\kappa}$ (300 paths)
- ▣ (c) Sample ACF of x_t at lags 1, 5, 20, 60, 120 against $\hat{b}^k$
- ▣ Model: B4

Interpretation

- ▣ Why is the bias much less important here than for interest rates?
- ▣ What does the ACF pattern say about a one-factor OU model?



B6: Merton and a Bootstrap Likelihood-Ratio Test [Solved]

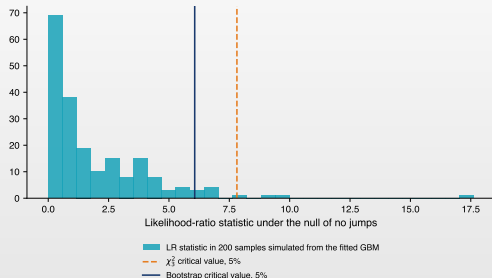
Task

- ▣ S&P 500 daily log returns 1990–2026 [Merton \(1976\)](#)
- ▣ (a) MLE of $(m, \sigma, \lambda, \mu_J, \sigma_J)$ with Hessian standard errors
- ▣ (b) Likelihood-ratio (LR) statistic against GBM
- ▣ (c) Its null distribution by parametric bootstrap: 200 GBM samples of the same length, both models refitted

Interpretation

- ▣ Why is the χ^2_3 distribution not valid for this LR test?
- ▣ Does the fitted model match the kurtosis and skewness of the data?

B6: Results [Solved]



- $\hat{\sigma} = 9.46\% (0.32)$, $\hat{\lambda} = 101.5 (10.5)$ per year, $\hat{\mu}_J = -0.162\% (0.034)$, $\hat{\sigma}_J = 1.45\% (0.06)$
 - Implied skewness -0.31 (data -0.36), excess kurtosis 3.71 (data 10.9)
- LR = 2397; bootstrap 95% critical value 6.07, against χ^2_3 : 7.81; bootstrap p-value 0 (none of 200 samples)
 - Under the null $\lambda = 0$ lies on the boundary and μ_J, σ_J are not identified: the usual asymptotics fail
- Jumps are overwhelmingly supported, yet the fit misses two thirds of the kurtosis

B7: Jumps in Bitcoin [Proposed]

Solution discussed in class

Task

- Bitcoin daily log returns 2014–2026, $\Delta t = 1/365$
- (a) Merton MLE with standard errors; implied excess kurtosis
- (b) Lee–Mykland test at 1% ($K = 16$) [Lee & Mykland \(2008\)](#): number of jumps per year and their mean size
- Model: B6

Interpretation

- Why do $\hat{\lambda}$ and the Lee–Mykland count differ by two orders of magnitude?
- Which estimate would you use to price a one-week Bitcoin option?

B8: Heston Parameters from the VIX [Proposed]

Solution discussed in class

Task

- ▣ S&P 500 and VIX, 1990–2026; $v_t = (\text{VIX}_t/100)^2$
Heston (1993)
- ▣ (a) Discretised CIR regression for κ, θ, ξ ; OLS standard error and 95% CI for κ
- ▣ (b) ρ with a Fisher-z 95% CI; the Feller ratio
- ▣ (c) Split the sample in 1990–2007 and 2008–2026
- ▣ Model: B4, B6

Interpretation

- ▣ Why is $\sqrt{\theta}$ from the VIX above the realised volatility?
- ▣ Are the parameters stable? Is the Feller condition met in both halves?

C1: Which Continuous-Time Model Fits BET and Bitcoin? [Proposed]

Solution discussed in class

- Fit GBM, Merton and Heston to BET and Bitcoin daily log returns
 - ▶ No VIX exists for these markets: use a 21-day realised variance as the variance proxy for Heston
- Simulate 100 samples of the same length from each fitted model
 - ▶ Compare excess kurtosis, Hill tail index, skewness and the ACF of $|r_t|$ with the data; report the percentile of the data in each simulated distribution
- Which features does each model reproduce, and which none of them?
- Project extension: add jumps to Heston (Bates) or use a GARCH-diffusion, and repeat; relate to the classification of crypto-assets by return distributions [Pele et al. \(2023\)](#)

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

The student asks an AI assistant

- "Assume the S&P 500 follows a GBM with $\mu = 8\%$ and $\sigma = 20\%$ per year. What is the probability that it ends the year below today's level? Check it by simulation."

The AI answer: text

- GBM: $dS = \mu S dt + \sigma S dW$
- By the chain rule, $d \ln S = dS/S = \mu dt + \sigma dW$, so $\ln(S_T/S_0) \sim N(\mu T, \sigma^2 T)$
- $P(S_T < S_0) = \Phi(-\mu\sqrt{T}/\sigma) = \Phi(-0.40) = 0.345$, with Φ the distribution function of the standard Normal distribution
- $\mathbb{E}[S_T] = S_0 e^{\mu T}$: on average the index ends the year 8.3% higher
- The simulation in the code gives 0.345 and confirms the formula

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np
mu, sigma, T, n, N = 0.08, 0.20, 1.0, 252, 20_000
dt = T / n
rng = np.random.default_rng(1)
Z = rng.standard_normal((N, n))
lnS = np.cumsum((mu - 0.5*sigma**2)*dt + sigma*dt*Z,
                axis=1)
print((lnS[:, -1] < 0).mean())
```

C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

Task

- ▣ The answer contains three errors; everything else is correct: find them
- ▣ For each error, say how you would detect it: a test, a published number, the lecture
- ▣ Give the corrected version: formula, code line and number
- ▣ Log each error as in `AI_ERRORS.md`
 - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

Hints

- ▣ Apply Itô's lemma to $f(S) = \ln S$ (A1): which term does the ordinary chain rule miss?
- ▣ Brownian increments over Δt have variance Δt : what is their standard deviation?
- ▣ Run the code: what does it actually print?

Common Pitfalls

- ▣ Forgetting the $-\sigma^2/2$ term: confusing the drift of the price with the drift of the log price
- ▣ Reporting a drift estimate without its standard error $\sigma/\sqrt{T}$
- ▣ Mixing units: annual parameters with daily Δt , trading days with calendar days (Bitcoin)
- ▣ Reading $\hat{\kappa}$ from a short sample as fast mean reversion: it is biased upwards
- ▣ Using χ^2 critical values for tests with a parameter on the boundary
- ▣ Treating the VIX as the real-world variance: it contains a risk premium

Take-Home Assignment (Team Project)

- ▣ Choose one asset not covered in class (a BVB stock, a currency or a crypto-asset)
- ▣ Fit GBM, Merton and a stochastic-volatility model; report every parameter with a standard error or a bootstrap CI
- ▣ Test jumps with Lee–Mykland and compare the jump count with the Merton intensity
- ▣ Simulate each model and rank them by how well they reproduce four stylised facts
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`

Further Study on Quantinar

Advanced courses on quantinar.com

-  Applied Computational Finance
-  XFG Advanced Methods in Quantitative Finance

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

References

- ▣ Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- ▣ Cont, R. (2001). *Empirical properties of asset returns: stylized facts and statistical issues*. *Quantitative Finance*, 1(2), 223–236.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Heston, S.L. (1993). *A closed-form solution for options with stochastic volatility with applications to bond and currency options*. *Review of Financial Studies*, 6(2), 327–343.
- ▣ Higham, D.J. (2001). *An algorithmic introduction to numerical simulation of stochastic differential equations*. *SIAM Review*, 43(3), 525–546.
- ▣ Lee, S.S., Mykland, P.A. (2008). *Jumps in financial markets: a new nonparametric test and jump dynamics*. *Review of Financial Studies*, 21(6), 2535–2563.
- ▣ Merton, R.C. (1976). *Option pricing when underlying stock returns are discontinuous*. *Journal of Financial Economics*, 3(1–2), 125–144.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). *Are cryptos becoming alternative assets?*. *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Tang, C.Y., Chen, S.X. (2009). *Parameter estimation and bias correction for diffusion processes*. *Journal of Econometrics*, 149(1), 65–81.
- ▣ Vasicek, O. (1977). *An equilibrium characterization of the term structure*. *Journal of Financial Economics*, 5(2), 177–188.